

MH540F1

PPS Compound | Glass Fiber Reinforced  
聚苯硫醚复合材料 | 玻璃纤维增强

Product Profile 产品概况

This product is a 40% glass fiber reinforced PPS compound with excellent strength, rigidity and processability.  
本产品是40%的玻璃纤维增强的聚苯硫醚复合材料，具有优异的强度、刚性和成型加工性能。

|                |                                                                                   |                             |                                                                 |                         |
|----------------|-----------------------------------------------------------------------------------|-----------------------------|-----------------------------------------------------------------|-------------------------|
| Color 颜色       | NC 本色, BK 黑色                                                                      | Component 成分 <sup>1</sup>   | PPS-GF40                                                        |                         |
| Form 形式        | Pellets 粒子                                                                        | Processing Method 加工方式      | Injection Molding 注塑成型                                          |                         |
| Status 状态      | Commercially Available 已商用                                                        | Yellow Card 黄卡 <sup>2</sup> | QMFZ2.E539835                                                   |                         |
| Features 特征    | High Temperature Resistance<br>Inherent Flame Retardancy                          | 耐高温<br>自阻燃                  | Excellent Chemical Resistance<br>Good Electrical Properties     | 优异的耐化学品性<br>良好的电气性能     |
| Application 用途 | Automotive & Transportation<br>Electrical & Electronics<br>Office & Household Use | 汽车与交通<br>电子电气<br>办公和家用      | Energy & Electricity<br>Industrial & Manufacturing<br>Aerospace | 能源与电力<br>工业和制造<br>航空和航天 |

Technical Data 技术参数

| Physical Properties 物理性质                                        | Value 数值 | Unit 单位           | Test Method 测试方法 |
|-----------------------------------------------------------------|----------|-------------------|------------------|
| Density<br>密度                                                   | 1.66     | g/cm <sup>3</sup> | ISO 1183-1       |
| Water Absorption, 23°C /24hrs.<br>吸水率, 23°C /24hrs.             | 0.02     | %                 | ISO 62           |
| Mold Shrinkage <sup>3</sup><br>成型收缩率 <sup>3</sup>               | 0.3/0.7  | %                 | ISO 294-4        |
| Mechanical Properties 机械性能                                      | Value 数值 | Unit 单位           | Test Method 测试方法 |
| Tensile Strength (5mm/min), 23°C<br>拉伸强度, 23°C                  | 190      | MPa               | ISO 527-1/-2     |
| Tensile Modulus (1mm/min), 23°C<br>拉伸模量, 23°C                   | 19       | GPa               | ISO 527-1/-2     |
| Tensile Strain at Break (5mm/min), 23°C<br>拉伸断裂应变, 23°C         | 1.3      | %                 | ISO 527-1/-2     |
| Flexural Strength (2mm/min), 23°C<br>弯曲强度, 23°C                 | 285      | MPa               | ISO 178          |
| Flexural Modulus (2mm/min), 23°C<br>弯曲模量, 23°C                  | 15.0     | GPa               | ISO 178          |
| Charpy Impact Strength, Notched, 23°C<br>简支梁冲击强度, 缺口, 23°C      | 11       | kJ/m <sup>2</sup> | ISO 179/1eA      |
| Charpy Impact Strength, Unnotched, 23°C<br>简支梁冲击强度, 无缺口, 23°C   | 58       | kJ/m <sup>2</sup> | ISO 179/1eU      |
| Charpy Impact Strength, Notched, -30°C<br>简支梁冲击强度, 缺口, -30°C    | 11       | kJ/m <sup>2</sup> | ISO 179/1eA      |
| Charpy Impact Strength, Unnotched, -30°C<br>简支梁冲击强度, 无缺口, -30°C | 55       | kJ/m <sup>2</sup> | ISO 179/1eU      |

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|                                                                                                |                  |                      |                         |
|------------------------------------------------------------------------------------------------|------------------|----------------------|-------------------------|
| Rockwell Hardness, R Scale<br>洛氏硬度, R标尺                                                        | 120              | -                    | ISO 2039-2              |
| Rockwell Hardness, M Scale<br>洛氏硬度, M标尺                                                        | 100              | -                    | ISO 2039-2              |
| <b>Thermal Properties 热性能</b>                                                                  | <b>Value 数值</b>  | <b>Unit 单位</b>       | <b>Test Method 测试方法</b> |
| Deflection Temp. Under Load (DTUL), 1.80MPa<br>载荷下热变形温度, 1.80MPa                               | 265              | °C                   | ISO 75-1/-2             |
| Deflection Temp. Under Load (DTUL), 0.45MPa<br>载荷下热变形温度, 0.45MPa                               | >270             | °C                   | ISO 75-1/-2             |
| Melting Temperature, (10°C/min)<br>熔融温度, (10°C/min)                                            | 282              | °C                   | ISO 11357-1/-3          |
| Coef. of Linear Thermal Expansion. <sup>3</sup> , -50~50°C<br>线性热膨胀系数 <sup>3</sup> , -50~50°C  | 1.5/4.0          | ×10 <sup>-5</sup> /K | ISO 11359-1/-2          |
| Coef. of Linear Thermal Expansion. <sup>3</sup> , 50~100°C<br>线性热膨胀系数 <sup>3</sup> , 50~100°C  | 1.5/6.0          | ×10 <sup>-5</sup> /K | ISO 11359-1/-2          |
| Coef. of Linear Thermal Expansion <sup>3</sup> , 100~200°C<br>线性热膨胀系数 <sup>3</sup> , 100~200°C | 1.5/8.5          | ×10 <sup>-5</sup> /K | ISO 11359-1/-2          |
| Flammability / Thickness (mm)<br>燃烧性 / 厚度 (mm)                                                 | V-0/0.75         | -                    | UL 94                   |
| Limiting oxygen index (LOI)<br>极限氧指数                                                           | 47               | %                    | ISO 4589                |
| <b>Electrical Properties 电性能</b>                                                               | <b>Value 数值</b>  | <b>Unit 单位</b>       | <b>Test Method 测试方法</b> |
| Electric Strength<br>介电强度                                                                      | 28               | kV/mm                | IEC 60243-1             |
| Relative Permittivity, 1MHz<br>相对介电常数, 1MHz                                                    | 3.6              | -                    | IEC 62631-2-1           |
| Dissipation Factor, 1MHz<br>介电损耗因数, 1MHz                                                       | 3.0              | ×10 <sup>-3</sup>    | IEC 62631-2-1           |
| Relative Permittivity, 2.5GHz/5GHz<br>相对介电常数, 2.5GHz/5GHz                                      | 3.9/4.0          | -                    | IEC 61189-2-721         |
| Dissipation Factor, 2.5GHz/5GHz<br>介电损耗因数, 2.5GHz/5GHz                                         | 4.8/5.5          | ×10 <sup>-3</sup>    | IEC 61189-2-721         |
| Comparative Tracking Index (CTI)<br>相对耐漏电起痕指数 (CTI)                                            | 175<br>(3)       | V<br>(PLC)           | IEC 60112<br>(UL746 A)  |
| Volume Resistivity<br>体积电阻率                                                                    | 10 <sup>14</sup> | Ω·m                  | IEC 62631-3-1           |
| Surface Resistivity<br>表面电阻率                                                                   | 10 <sup>16</sup> | Ω                    | IEC 62631-3-2           |

### Note 注释

1. In compliance with ISO 1043 符合ISO 1043标准
2. <https://iq.ulprospector.com/zh-cn/profile?e=7569313>
3. FD/TD, Flow Direction 流动方向 / Transverse Direction 垂直方向

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### Typical Injection Molding Conditions 典型注塑成型工艺

|                                                                                                                |                        |              |             |                       |              |             |            |
|----------------------------------------------------------------------------------------------------------------|------------------------|--------------|-------------|-----------------------|--------------|-------------|------------|
| Pre-drying 预干燥                                                                                                 |                        |              |             |                       |              |             |            |
| 120°C / 4 h; or 130°C / 3 h; or 140°C / 2 h<br>120°C / 4 小时; 或者 130°C / 3 小时; 或者 140°C / 2 小时                  |                        |              |             |                       |              |             |            |
| Mold Temp. 模具温度                                                                                                |                        |              |             |                       |              |             |            |
| 130 ~ 150°C (Recommended to use 140°C and avoid the 80~120°C range)<br>130 ~ 150°C (建议使用140°C, 并避开80~120°C的范围) |                        |              |             |                       |              |             |            |
| Injection Conditions 注塑工艺                                                                                      |                        |              |             |                       |              |             |            |
| Barrel Profile<br>料筒温度                                                                                         | Resin (Melt)<br>树脂(熔体) | Nozzle<br>喷嘴 | Zone4<br>四区 | Zone3<br>三区           | Zone2<br>二区  | Zone1<br>一区 | Feed<br>喂料 |
| Max (°C) 上限                                                                                                    | 340                    | 330          | 340         | 330                   | 320          | 300         | 100        |
| Rec. (°C) 推荐值                                                                                                  | 320                    | 315          | 320         | 315                   | 310          | 290         | -          |
| Min (°C) 下限                                                                                                    | 300                    | 300          | 300         | 300                   | 300          | 280         | -          |
| Injection Pressure<br>注射压力                                                                                     | 50 ~ 100 MPa           |              |             | Pack Pressure<br>保压压力 | 30 ~ 80 MPa  |             |            |
| Injection Rate<br>注射速率                                                                                         | Medium to Fast 中高速     |              |             | Pack Time<br>保压时间     | 3.0 ~ 10.0 s |             |            |
| Injection Time<br>注射时间                                                                                         | 0.5 ~ 2 s              |              |             | Back Pressure<br>背压   | 0.3 ~ 3 MPa  |             |            |
| Screw Cushion<br>射胶终点                                                                                          | 3 ~ 8 mm               |              |             | Screw Speed<br>螺杆转速   | 50 ~ 100 RPM |             |            |

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